

Network Topologies

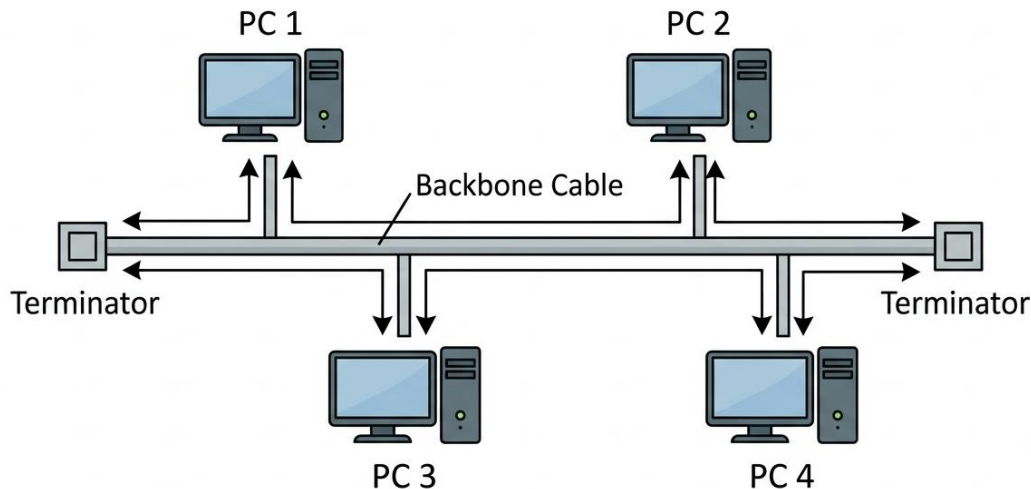
Task 1: Bus Topology

A bus topology is a network setup where all computers or devices are connected to one main cable called the backbone cable. The backbone cable carries the data from one device to another. Terminators are placed at both ends of the cable to prevent signal reflection.

Real-world example:

Bus topology can be used in a small temporary network or in some older Ethernet networks where multiple computers share one main communication cable.

Bus Network Topology



Star Topology in Cisco Packet Tracer

For this practical, I created a simple star topology using one switch and four PCs. All four PCs are directly connected to the central switch.

Devices Used:

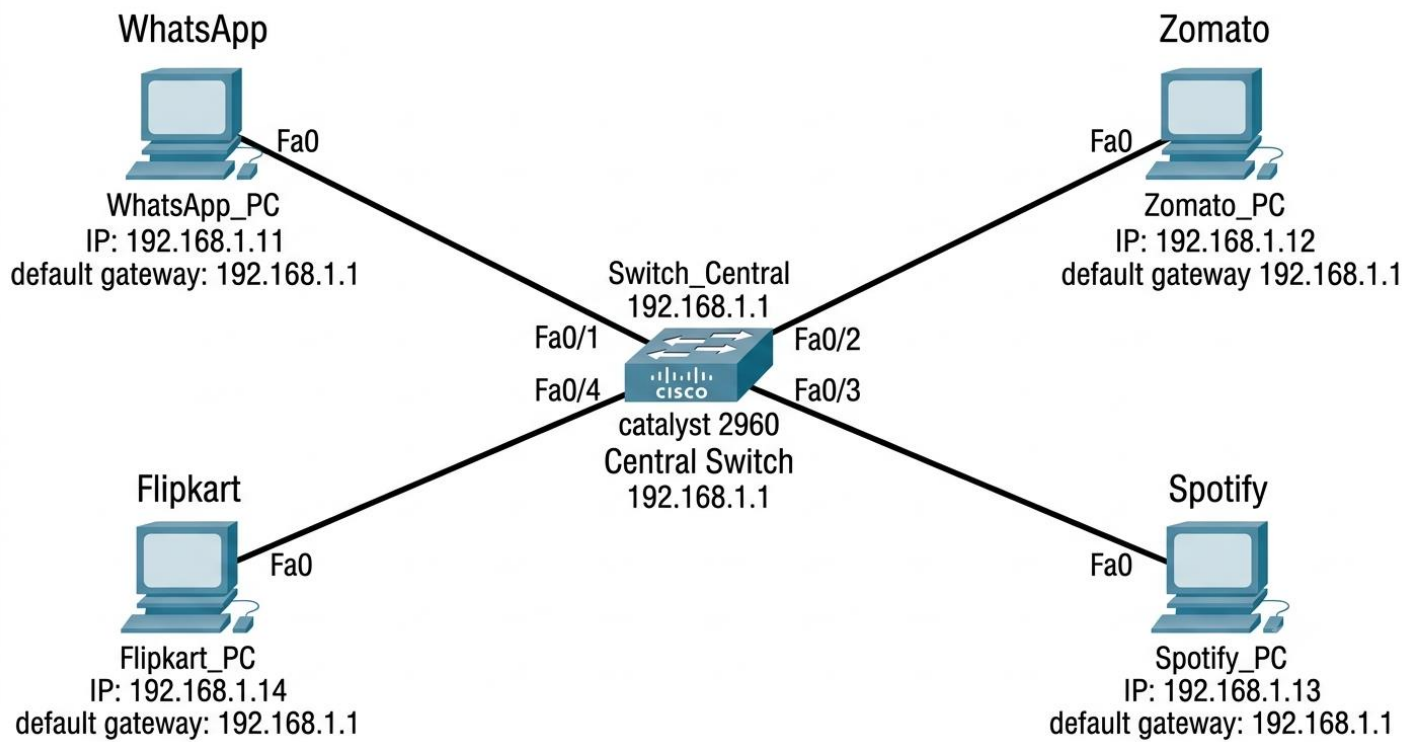
- 1 Switch
- 4 PCs
- 4 Copper Straight-Through cables

PC Names:

- PC0 → WhatsApp
- PC1 → Zomato
- PC2 → Spotify
- PC3 → Flipkart

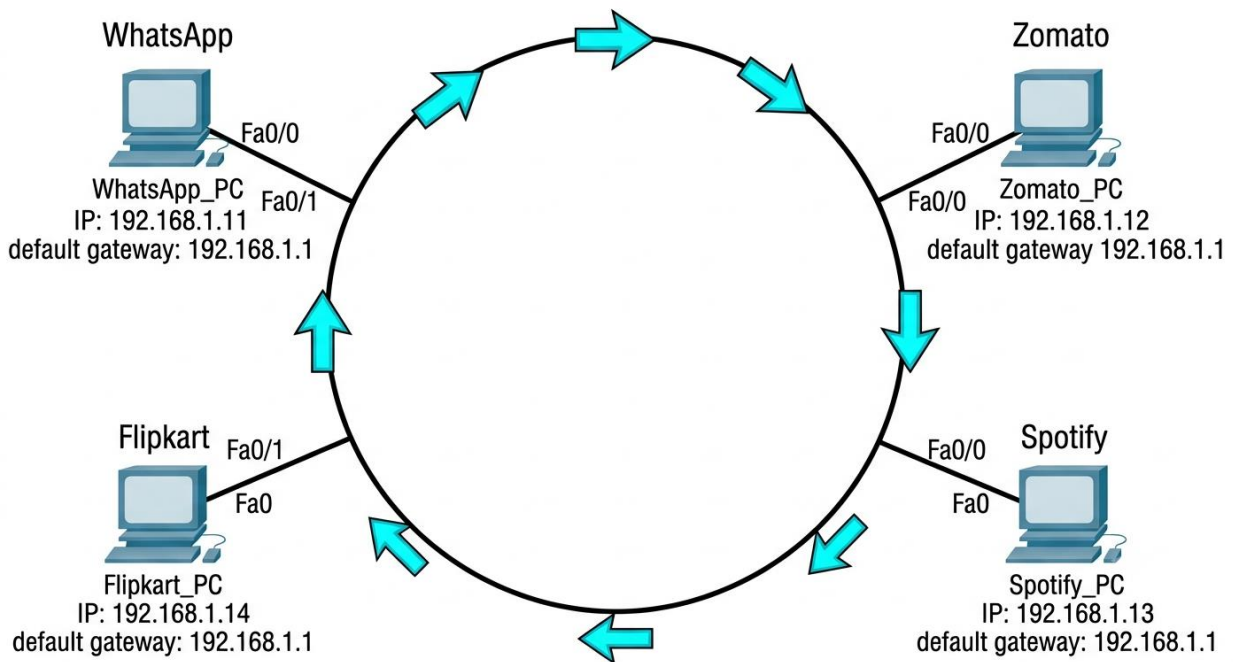
Steps:

1. Open Cisco Packet Tracer.
2. Add one switch to the workspace.
3. Add four PCs around the switch.
4. Connect each PC to the switch using Copper Straight-Through cable.
5. Rename the PCs as WhatsApp, Zomato, Spotify and Flipkart.
6. Arrange the devices around the switch so that the network looks like a star.
7. Save the Packet Tracer file.
8. Take a screenshot of the completed topology and add it below.



Ring Topology

A ring topology connects each device to two other devices and forms a closed circle. Data travels from one device to the next device around the ring until it reaches the destination. Arrows can be used to show the direction in which the data is travelling.



mesh vs Star Topology

Topology	Advantages	Disadvantages
Star	1. Easy to install and manage. 2. If one PC or cable fails, other devices can still work.	1. If the central switch fails, the whole network can stop. 2. Requires more cables than a bus topology.
Mesh	1. Provides multiple paths for data. 2. Very reliable because one connection failure does not normally stop the whole network.	1. Requires a lot of cables and connections. 2. It is more expensive and difficult to install and maintain.

Conclusion

In a star topology, all devices are connected to one central switch, which makes the network easy to manage. A mesh topology provides multiple connections between devices, so it gives better reliability but requires more cables and is more expensive.